

BHUVAN KODURU

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EDUCATION

Carnegie Mellon University, M.S. in AI/NLP (Intelligent Information Systems) Pittsburgh, PA | Dec 2026
GPA: **4.05/4.00** | Coursework: Multimodal Machine Learning, Speech Technology for Conversational AI, Deep Learning
JSS Science and Technology University, B.E. in Information Science & Engineering Mysore, India | Jun 2024
GPA: **9.44/10.00**

EXPERIENCE

Similar.ai — *Research Intern* Palo Alto, CA | Jun – Aug 2026

- **Co-authored CUES-Evol** (CoLM-W 2026) and Agent S4: enabled computer-use agents to self-generate their interface in code, matching specialized scaffolds on OSWorld (361 tasks) with no CUA-specific engineering.
- Built an **island-based evolutionary loop** growing a reusable-code repo from rollouts alone, beating Agent S3 on **ScienceBoard** (Haiku 30.9% vs. 22.0%, Sonnet 40.4% vs. 30.1%) at **half the token cost**.
- Demonstrated via **mechanistic interpretability** that chat safety does not transfer to agents across 8 models and 49 tasks: refusal collapsed from **31/47 (chat) to 4/47 (agent)**; activation-steering that raised chat refusal (0.71 to 0.85) left agent actions unchanged.
- Designed the study's **human-rating interface**—blinded trajectories with a few high-signal questions per item—to **minimize rater effort and maximize information per label** (13 raters, 363 ratings), framing the evaluation around the result that matters to its audience.

Carnegie Mellon University — *Graduate Research Assistant* Pittsburgh, PA | Jan – May 2026

- Built an evaluation harness simulating **8,000+ Among Us games** across **21 LLMs** to quantify deception and Theory of Mind.
- Released a novel benchmark dataset with a moderator agent grounded in a game-theoretic model of strategic deception.

Adobe Inc. — *Associate Technical Consultant* Bengaluru, India | Aug 2024 – Apr 2025

- Architected a **multi-agent pipeline** automating plaintext-to-Fusion conversion; advanced to **top-24 semifinalists worldwide**.
- Built a Workfront Fusion tool to flag orphan nodes and generate AI summaries; cut documentation time **70%**.

Nosh Robotics — *Computer Vision Intern* Bengaluru, India | Feb – Jun 2024

- Trained and quantized **ResNet, Inception, and MobileNet** classifiers for real-time edge estimation of ingredient cooking levels; models were **deployed to production**.
- Applied zero-shot **CLIP** for pan detection and built a RAG food Q&A chatbot integrating **FAISS** and **ChromaDB**.

Spinal Cord Lab, IIT Hyderabad — *Research Intern* Remote | Oct 2023 – Mar 2024

- Analyzed motion-capture and mobile-camera data across **17 subjects / 23 activities**, using LED-blink sequences for time synchronization.
- Diagnosed and resolved a critical pipeline bug affecting all experiments, then computed per-joint offsets to benchmark estimation accuracy.

Adobe Inc. — *Technical Consultant Intern* Bengaluru, India | May – Jul 2023

- Deployed an AI agent inside Workfront Fusion with @-mention commands calling **ChatGPT, Photoshop, and Stable Diffusion** APIs, reducing campaign turnaround from weeks to hours.

PUBLICATIONS

- Heard but Not Heeded: Paralinguistic Information Encoding and Loss in Audio-Language Models — **B. Koduru**, D. Alharthi, R. Singh, B. Raj. (**CoLM 2026**)
- RIVET: Robust Idempotent Voice Attribute Editing — D. Alharthi, **B. Koduru** et al. (**INTERSPEECH 2026**, Oral)
- Chat-safe is not CUA-safe: Chat Methods May Not Transfer to Computer Use — **B. Koduru**, J. Yang, A. Li, E. Wang (**Agents in the Wild Workshop @ NeurIPS 2026**, Poster)
- Agent S4: Evolutionary Computer Use — S. Agashe, Y. Yang, G. Gonzalez-Pumariiega, **B. Koduru** et al. (**ICLR 2027**, under review)
- Subliminal Prosody Learning: Auxiliary Emotion Supervision Redistributes Affective Representations Across ALM Layers — **B. Koduru** et al. (**ICML Mechanistic Interpretability Workshop 2026**)
- Computer Use with Evolving Software — S. Agashe, G. Gonzalez-Pumariiega, **B. Koduru** et al. (**Lifelong Agents Workshop @ CoLM 2026**)
- AmongUs-X: Benchmarking Strategic Deception of LLM Agents via Theory-of-Mind Metrics — K. Mirakhor*, **B. Koduru*** et al. (**AAMAS 2027**, under review)

PROJECTS

- **Persona-Induced Response Modulation** — Engineered an activation-steering pipeline inducing confident personas without fine-tuning; achieved **95% uncertainty reduction**, ablated on Llama 3.1 8B and Gemma 2 2B. (*Oct – Dec 2025*)
- **Arabic TTS Fine-Tuning** — Supervised fine-tuning of NeuTTS-Air on a proprietary Arabic dataset achieving **30% WER** in Prof. Bhiksha Raj's MLSP lab. (*Sep – Dec 2025*)

SKILLS

LLMs & Agents: mechanistic interpretability, activation steering, multi-agent systems, RAG, vLLM, ChromaDB
ML / DL: fine-tuning (LoRA/SFT), PyTorch, Hugging Face Transformers, quantization, NumPy, Pandas
Languages & Tools: Python, C/C++, SQL, Git, Linux, Docker, CUDA, MongoDB, Flask